

Credit-Market Sentiment and the Business Cycle: A Replication of López-Salido, Stein & Zakrajšek (2017)

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Abstract

We replicate the core results of López-Salido, Stein, and Zakrajšek (2017), which shows that elevated credit-market sentiment—“frothy” credit conditions—forecasts a slowdown in real activity roughly two years ahead. This document describes the project at a high level, explains the data sources, reviews where the replication succeeded and where it was difficult, and collects every figure and table our code produces across three sets: the original replication sample, a sample extended to the present, and an Aaa-spread extension.

1 Overview of the Project

The paper’s central claim is that the mood of credit markets carries information about the future path of the business cycle. When the Baa–Treasury spread is unusually narrow and a large share of new issuance is low-quality, investors are pricing credit risk aggressively; those conditions mean-revert, and spreads widen back out as activity turns down. Our code reproduces the paper’s key exhibits—the long-run credit spread, the predictive regressions of output growth on credit variables, and the sentiment measure built from the Greenwood–Hanson high-yield issuance share—and re-runs each on data extended to the present and on an Aaa-based variant.

2 Data Sources

Every series is pulled programmatically so the results are reproducible:

- **Federal Reserve (FRED).** Moody’s seasoned Baa and Aaa corporate bond yields, the 10-year Treasury yield, real GDP, and related macroeconomic series, forming the credit spreads and output outcomes. Some of these series are only available in a newer vintage than the one used in the original paper, and several require stitching together multiple FRED series to obtain a continuous long history. The paper’s appendix does not always specify exactly which underlying series were used, so where the mapping was ambiguous we made a documented judgement call on which series to combine; this is a further source of small differences from the published numbers.
- **Robert Shiller’s website.** Long-run stock-market and interest-rate data, including the cyclically adjusted price-earnings ratio used as the equity-return predictor.
- **Greenwood & Hanson high-yield issuance share.** The share of high-yield issuance in total corporate bond issuance, underlying the credit-market sentiment measure. The historical portion (1926–2015) is pulled directly from its published source, the HBS Behavioral

Finance and Financial Stability `InvestorCreditSentiment` workbook, and spliced with a reconstruction from Mergent FISD (via WRDS) for recent years. The HBS workbook is a newer vintage than the values printed in the original paper’s table, so some replicated numbers differ slightly from the published ones; we use the updated vintage deliberately, as it reflects the authors’ latest revisions.

3 Replication Sample (1929–2015)

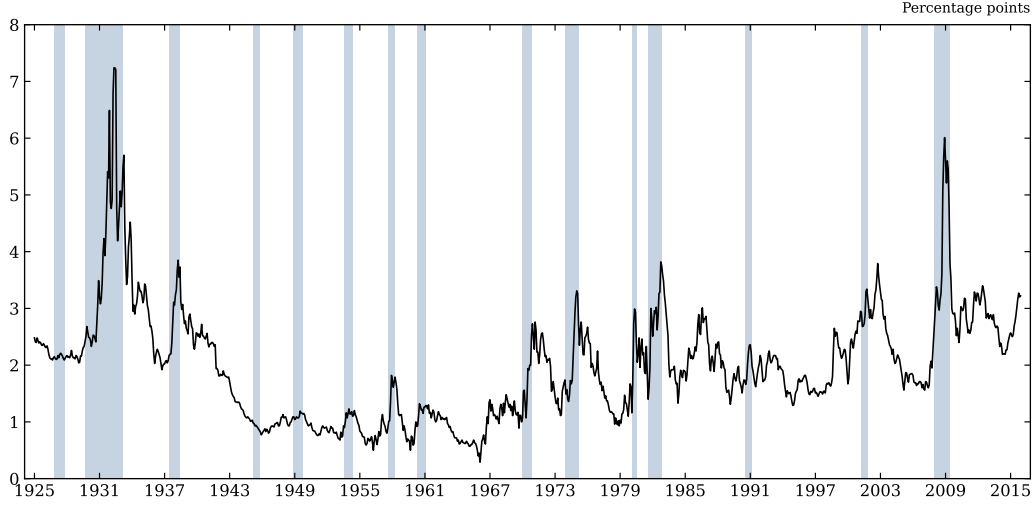


Figure 1: Figure I (replication): the Baa–Treasury credit spread, NBER recessions shaded.

Table I (replication).

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.958*** (0.585)	—	-2.124*** (0.578)
r_t^{SP}	—	0.075** (0.033)	0.022 (0.035)
Δy_{t-1}	0.476*** (0.104)	0.478*** (0.136)	0.464*** (0.114)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.248 (0.290)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.814** (0.360)
π_{t-1}	—	—	0.095 (0.076)
$\bar{R}^2$	0.426	0.381	0.453
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.365	—	-0.396
r_t^{SP}	—	0.299	0.087

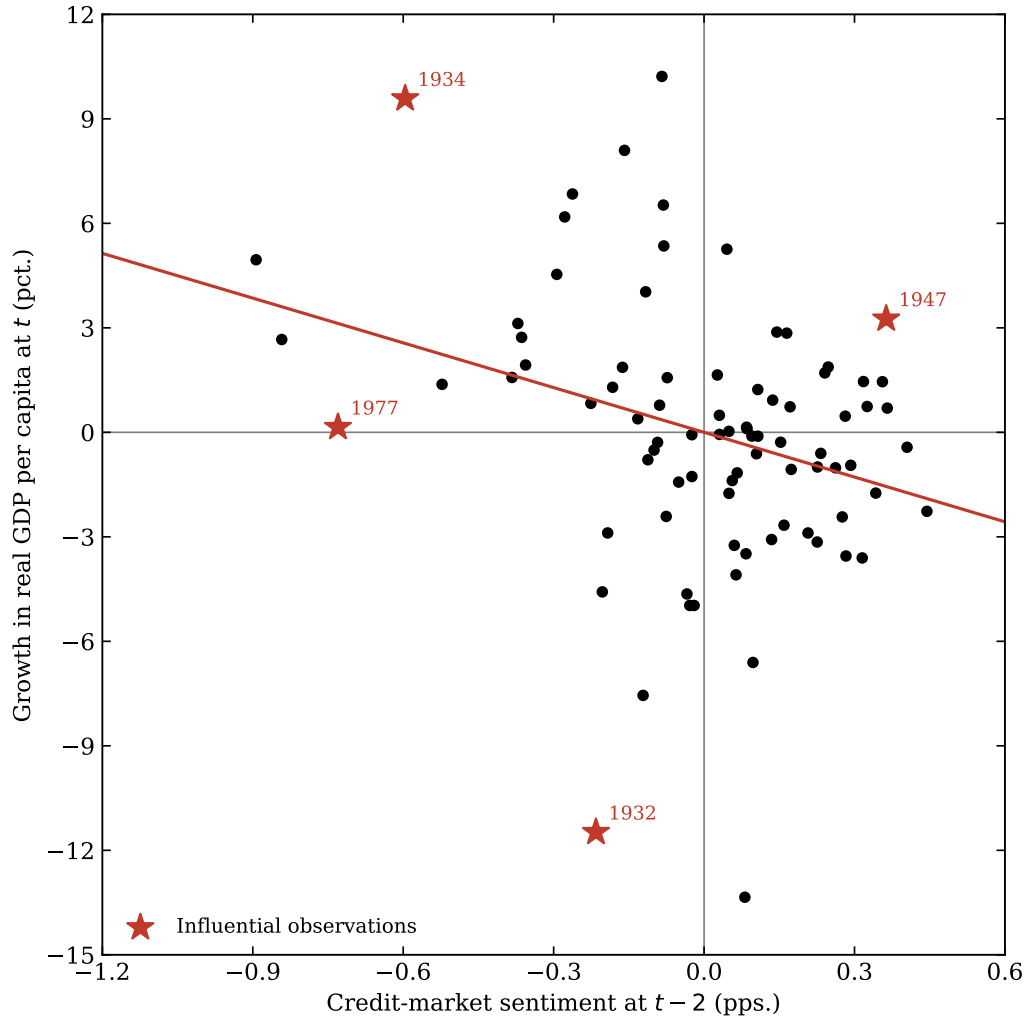


Figure 2: Figure II (replication): the credit-market sentiment measure.

Table II (replication).

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.281*** (1.253)	—	-3.861*** (1.301)	-4.907*** (1.678)
$\hat{r}_t^{SP}$	—	0.145* (0.082)	0.053 (0.074)	—
Δy_{t-1}	0.597*** (0.114)	0.534*** (0.099)	0.590*** (0.112)	0.581*** (0.096)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.099 (0.313)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.616 (0.414)
π_{t-1}	—	—	—	0.123 (0.154)
R^2	0.377	0.337	0.380	0.394
Auxiliary regressions				
		Δs_t	r_t^{SP}	
$\ln \text{HYS}_{t-2}$		0.147*** (0.042)	—	
s_{t-2}		-0.245*** (0.047)	—	
$\ln[P/E10]_{t-2}$		—	-0.132*** (0.040)	
R^2		0.102	0.083	

4 Extended Sample (1929–2025)

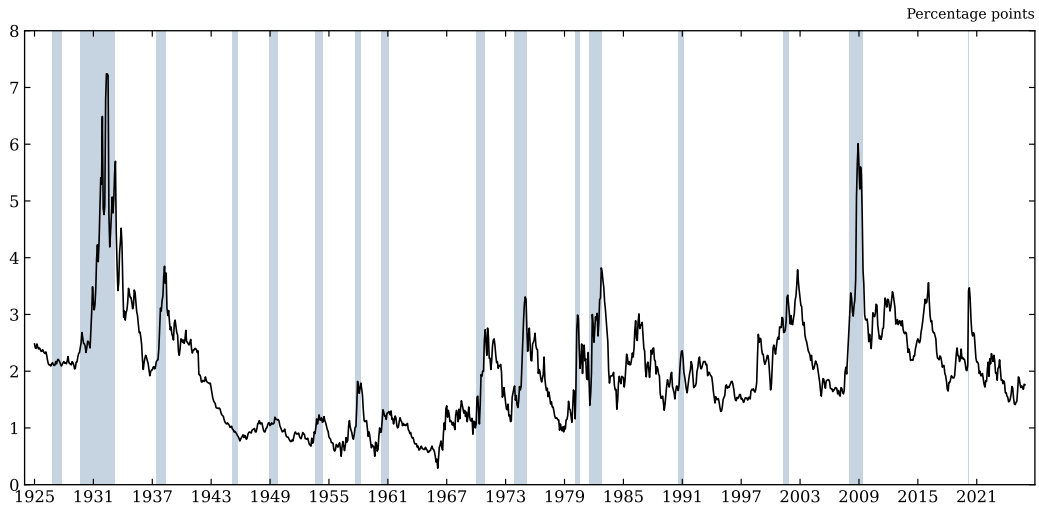


Figure 3: Figure I extended to the present.

Table I (extended).

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.874*** (0.566)	—	-2.020*** (0.572)
r_t^{SP}	—	0.070** (0.033)	0.018 (0.035)
Δy_{t-1}	0.461*** (0.108)	0.464*** (0.136)	0.450*** (0.116)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.183 (0.269)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.752** (0.367)
π_{t-1}	—	—	0.088 (0.078)
$\bar{R}^2$	0.395	0.354	0.415
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.350	—	-0.378
r_t^{SP}	—	0.286	0.072

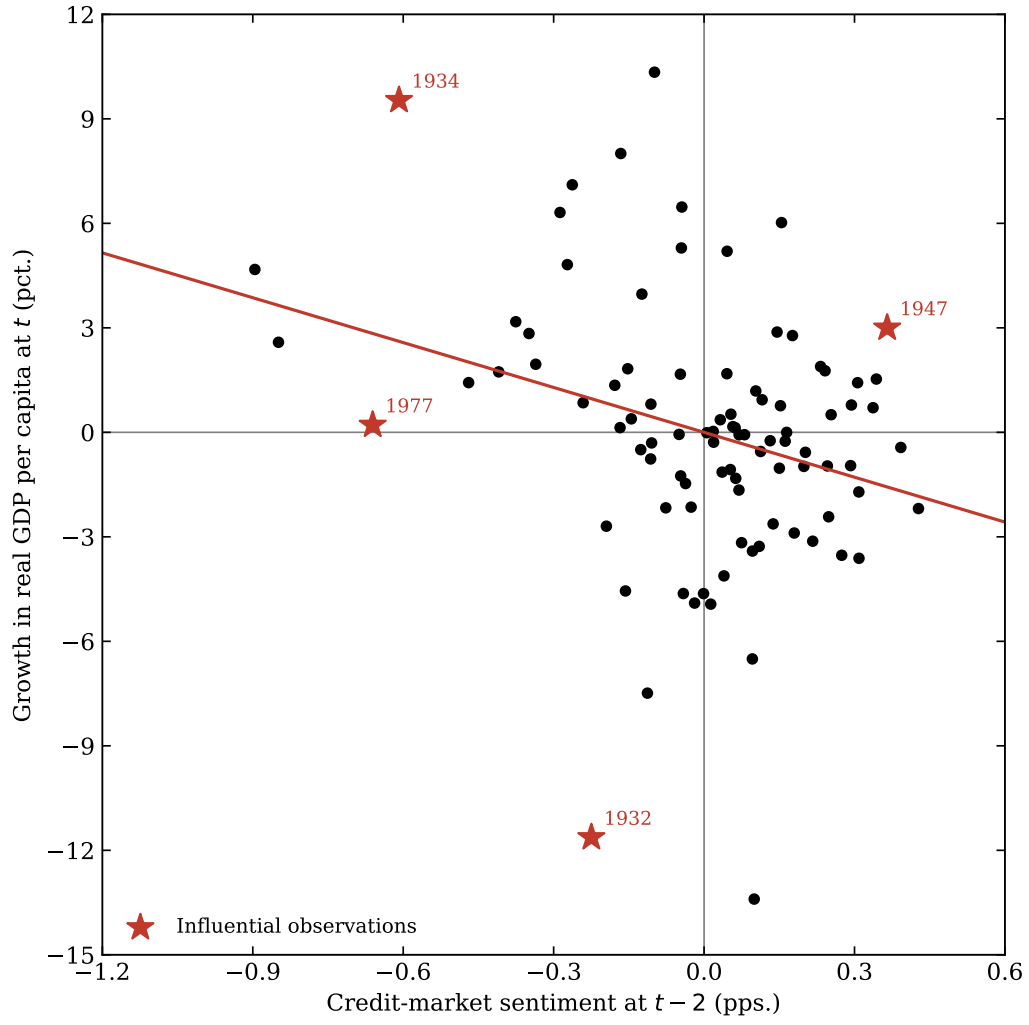


Figure 4: Figure II extended to the present.

Table II (extended).

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-4.294*** (1.295)	—	-3.901*** (1.316)	-4.943*** (1.703)
$\hat{r}_t^{SP}$	—	0.158* (0.092)	0.062 (0.081)	—
Δy_{t-1}	0.577*** (0.118)	0.516*** (0.103)	0.570*** (0.115)	0.559*** (0.100)
$\Delta i_{t-1}^{(3m)}$	—	—	—	0.123 (0.290)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.612 (0.410)
π_{t-1}	—	—	—	0.121 (0.154)
R^2	0.353	0.313	0.356	0.368
Auxiliary regressions				
		Δs_t	r_t^{SP}	
$\ln \text{HYS}_{t-2}$		0.129*** (0.039)	—	
s_{t-2}		-0.243*** (0.046)	—	
$\ln[P/E10]_{t-2}$		—	-0.103*** (0.034)	
R^2		0.095	0.059	

5 Extension: Aaa Credit Spread

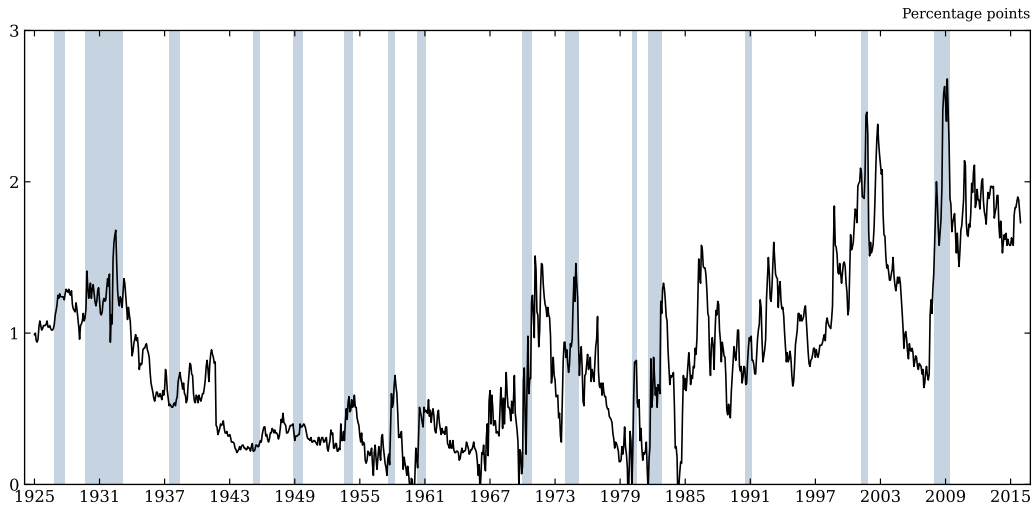


Figure 5: Figure I, Aaa spread (replication sample).

Table I, Aaa (replication).

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.958*** (0.734)	—	-3.192*** (1.168)
r_t^{SP}	—	0.075** (0.033)	0.056 (0.035)
Δy_{t-1}	0.517*** (0.109)	0.478*** (0.136)	0.477*** (0.121)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.326 (0.314)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.917* (0.504)
π_{t-1}	—	—	0.102 (0.108)
$\bar{R}^2$	0.321	0.381	0.396
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.165	—	-0.269
r_t^{SP}	—	0.299	0.225

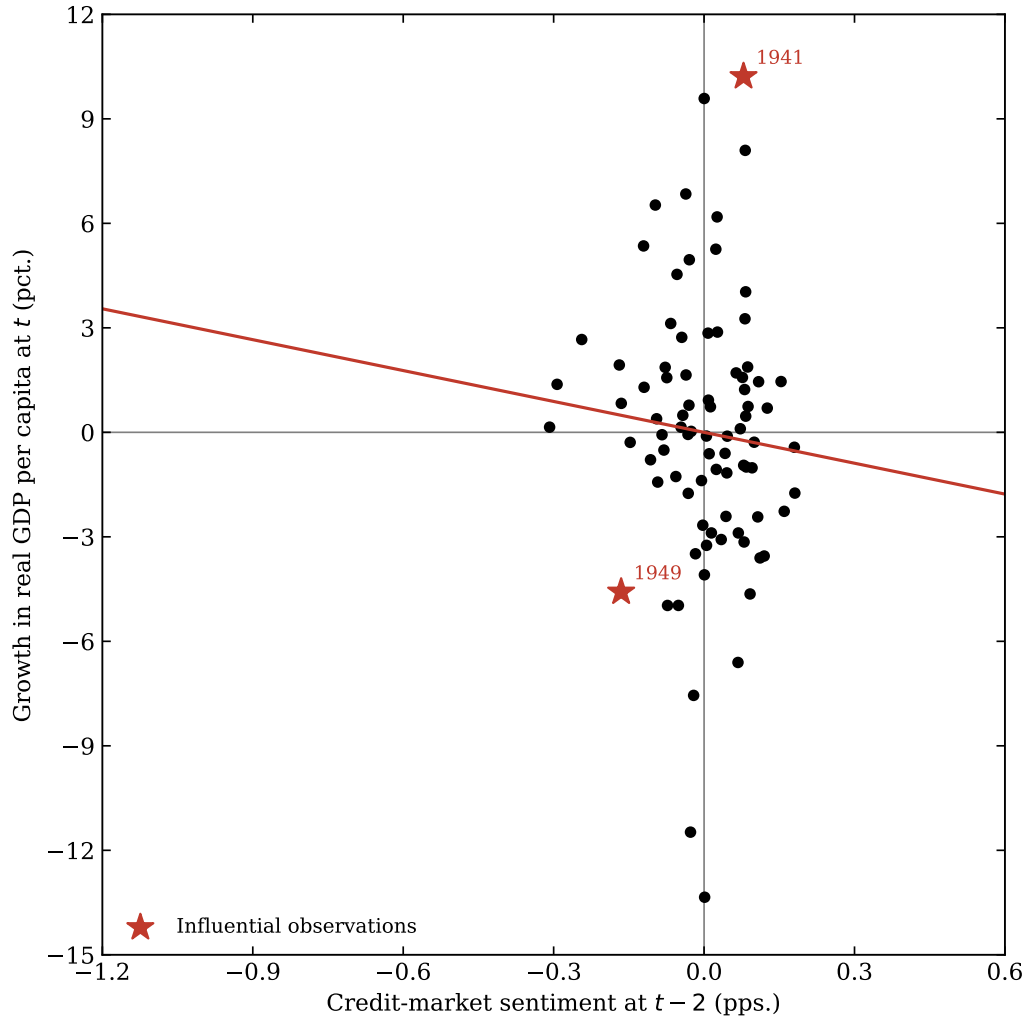


Figure 6: Figure II, Aaa spread (replication sample).

Table II, Aaa (replication).

Dependent variable: Δy_t				
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.956 (2.488)	—	-2.362 (2.865)	-3.663 (2.806)
$\hat{r}_t^{SP}$	—	0.145* (0.082)	0.131 (0.085)	—
Δy_{t-1}	0.555*** (0.108)	0.534*** (0.099)	0.549*** (0.100)	0.559*** (0.105)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.063 (0.308)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.373 (0.394)
π_{t-1}	—	—	—	0.043 (0.132)
R^2	0.321	0.337	0.342	0.328
Auxiliary regressions	Δs_t		r_t^{SP}	
$\ln \text{HYS}_{t-2}$	0.078*** (0.029)		—	
s_{t-2}	-0.167*** (0.058)		—	
$\ln[P/E10]_{t-2}$	—		-0.132*** (0.040)	
R^2	0.064		0.083	

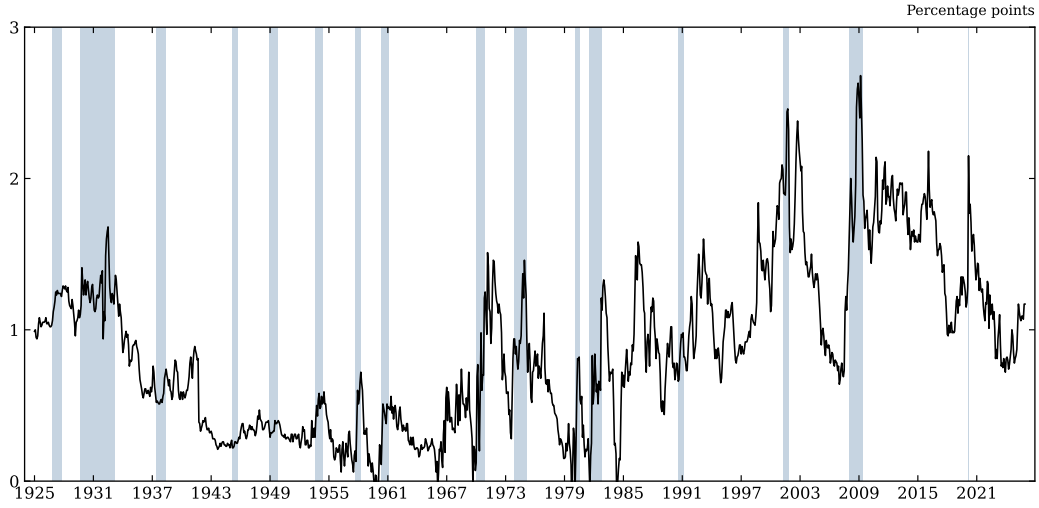


Figure 7: Figure I, Aaa spread (extended sample).

Table I, Aaa (extended).

Regressors	Dependent variable: Δy_t		
	(1)	(2)	(3)
Δs_{t-1}	-1.810** (0.714)	—	-2.873** (1.165)
r_t^{SP}	—	0.070** (0.033)	0.052 (0.035)
Δy_{t-1}	0.500*** (0.112)	0.464*** (0.136)	0.463*** (0.124)
$\Delta i_{t-1}^{(3m)}$	—	—	-0.257 (0.289)
$\Delta i_{t-1}^{(10y)}$	—	—	-0.797 (0.495)
π_{t-1}	—	—	0.091 (0.109)
$\bar{R}^2$	0.297	0.354	0.360
<i>Standardized effect on Δy_t</i>			
Δs_{t-1}	-0.154	—	-0.244
r_t^{SP}	—	0.286	0.210

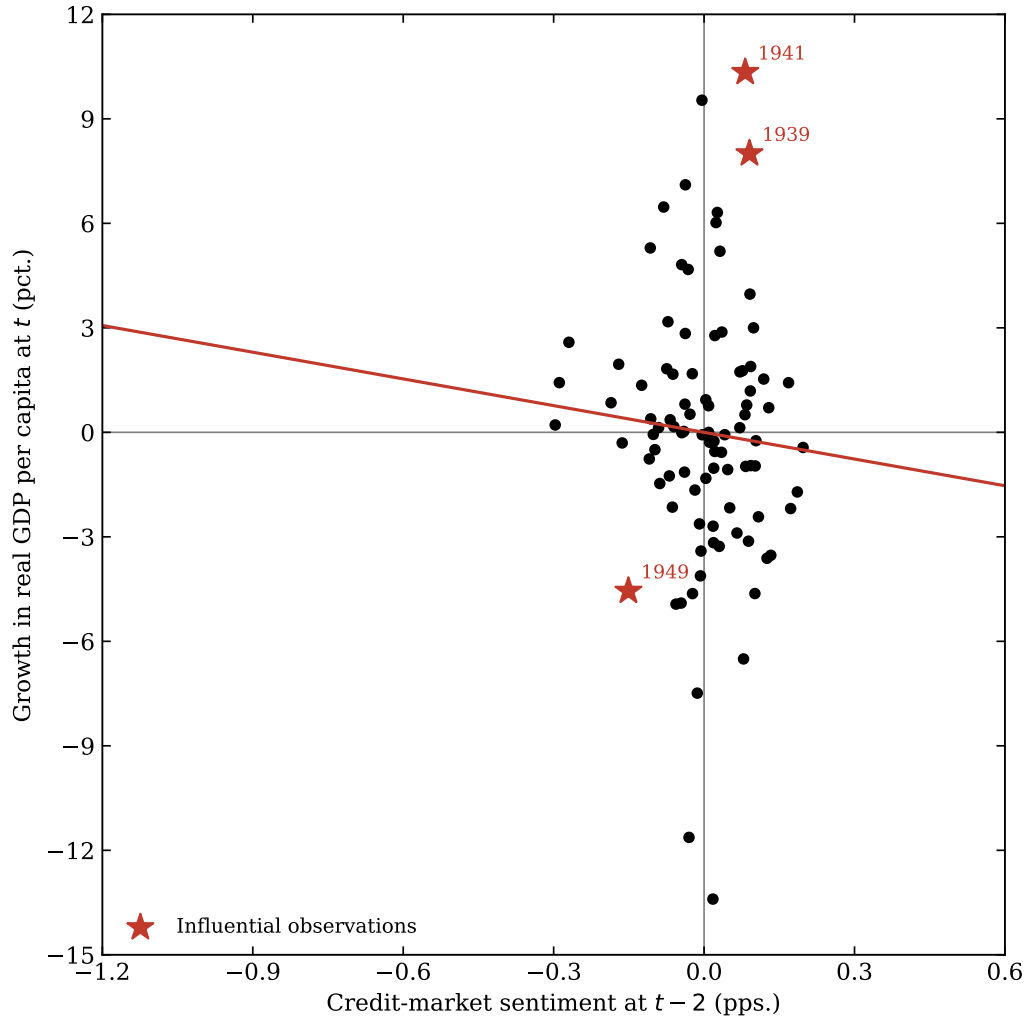


Figure 8: Figure II, Aaa spread (extended sample).

Table II, Aaa (extended).

	Dependent variable: Δy_t			
	(1)	(2)	(3)	(4)
$\Delta \hat{s}_t$	-2.556 (2.389)	—	-2.546 (2.578)	-3.219 (2.717)
$\hat{r}_t^{SP}$	—	0.158* (0.092)	0.146 (0.094)	—
Δy_{t-1}	0.536*** (0.111)	0.516*** (0.103)	0.532*** (0.103)	0.540*** (0.107)
$\Delta i_{t-1}^{(3m)}$	—	—	—	-0.050 (0.286)
$\Delta i_{t-1}^{(10y)}$	—	—	—	-0.367 (0.389)
π_{t-1}	—	—	—	0.041 (0.132)
R^2	0.298	0.313	0.318	0.306
Auxiliary regressions		Δs_t	r_t^{SP}	
$\ln \text{HYS}_{t-2}$		0.076*** (0.028)	—	
s_{t-2}		-0.182*** (0.058)	—	
$\ln[P/E10]_{t-2}$		—	-0.103*** (0.034)	
R^2		0.069	0.059	

6 Successes and Challenges

Where it worked. The long-run credit spread (Figure I) reproduces cleanly from FRED and shows the expected countercyclical spikes into every recession. The central result held in Table II: changes in the fitted credit spread forecast subsequent output growth (main specification $R^2 \approx 0.38$), and the auxiliary regression confirms the lagged log high-yield share and lagged spread jointly predict spread changes. Extending the sample to the present preserved the qualitative result.

Where it was hard. The main difficulty was the Greenwood–Hanson high-yield issuance share, which is not a clean FRED pull; sourcing it reliably was the chief data bottleneck, resolved by pulling the published workbook directly from its URL and splicing it with a Mergent FISD reconstruction via WRDS. Assembling the FRED series was a second source of friction: several series had to be stitched together to form a continuous long history, and the paper’s appendix does not fully pin down which underlying series were used, so some choices were our own documented judgement calls. Matching the paper’s inference also required Newey–West standard errors with a suitable lag, and differences in sample period and data vintage shift point estimates relative to the published tables—expected, and explained by the updated data vintages and series choices noted above.